

Chapter 4

Lab 4 - Resistive Circuits

4.1 Objective

The objective of the laboratory is to introduce basic resistive circuits, the effect of resistors in series and parallel, applications of simple resistor ladders, and the effects of real world resistors.

4.2 Materials

- Laptop with Waveforms
- Digilent Analog Discovery
- Breadboard
- Wiring kit
- Lab parts kit

4.3 Introduction

Resistors are a fundamental part of all electronic circuits. Understanding how they work and how to apply them is a basic requirement for any electronics course.

4.3.1 Resistor Combinations

Resistors in series will always add together:

$$R_{Total} = R_1 + R_2 \dots R_N \quad (4.1)$$

while resistors in parallel combine as the addition of reciprocals:

$$\frac{1}{R_{Total}} = \frac{1}{R_1} + \frac{1}{R_2} \cdots \frac{1}{R_N} \quad (4.2)$$

Series resistances are straight forward, add one resistor to the other, but the parallel combination leads to different results based on the values of the resistors in parallel. For instance, a large resistor in parallel with a small resistor gives an equivalent resistance approximately equal to the small resistor. Similarly, two resistors of the same value in parallel gives an equivalent resistance equal to half of either resistor.

4.3.2 Voltage and Current Dividers

Resistors are commonly used to create voltage and current dividers, with the choice depending on the application. They are often employed to generate a smaller voltage from a larger voltage or to produce a smaller current from a larger current.

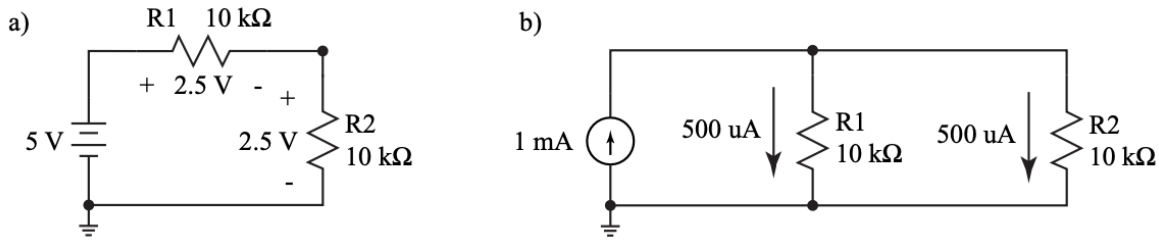


Figure 4.1: Examples of a voltage divider (a) and a current divider (b)

The output of a voltage divider can be simply calculated as:

$$V_{out} = \frac{R_2}{R_1 + R_2} V_{in} \quad (4.3)$$

In the example shown in (a), the output voltage is 2.5 V, which corresponds to the voltage across R2. This result is obtained by applying the voltage divider equation. Since the values for R1 and R2 are equal, the voltage drop across R1 is also 2.5 V.

$$\left(\frac{10 \text{ k}\Omega}{10 \text{ k}\Omega + 10 \text{ k}\Omega} \right) 5 \text{ V} = 2.5 \text{ V} \quad (4.4)$$

Similarly, a current divider is calculated in the same way where:

$$I_{out} = \frac{R2}{R1 + R2} I_{in} \quad (4.5)$$

and the resulting current in each resistor is 500 μ A which is again found by plugging in to the equation. Because the values for R1 and R2 are the same, the current through R1 is also 500 μ A.

$$\left(\frac{10 \text{ k}\Omega}{10 \text{ k}\Omega + 10 \text{ k}\Omega} \right) 1 \text{ mA} = 500 \text{ }\mu\text{A} \quad (4.6)$$

Note that a voltage divider applies to resistors in series while a current divider applies to resistors in parallel.

4.3.3 Resistor Ladders

Resistors are often used in a string or ladder configuration to generate reference voltages with a specific spacing. Typical applications of a resistor ladder are analog to digital conversion where the input voltage is compared to voltages on the resistor ladder or digital to analog conversion where digital bits (1s and 0s) are converted to an analog voltage.

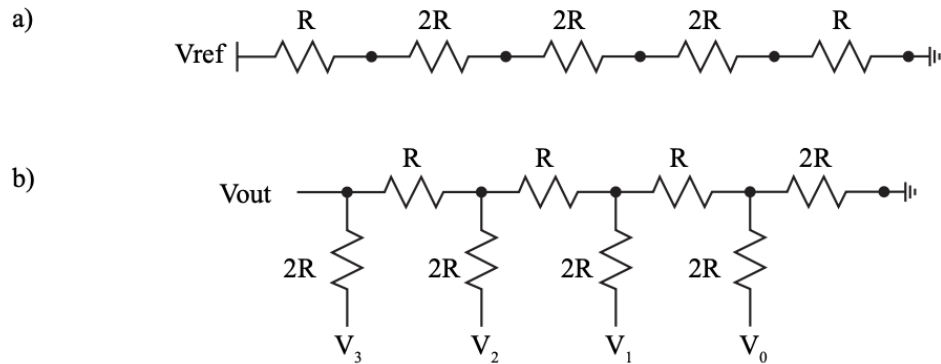


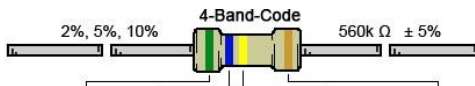
Figure 4.2: A simple resistor ladder where the resistor values set the voltages at each node (a) and a 4-bit R-2R ladder where digital bits, represented as voltages V_0 through V_3 are converted to an analog voltage, V_{out} .

4.3.4 Measuring Voltage and Current

In the past, separate meters were required to measure voltages and currents. Today, however, analog meters have been replaced by digital multimeters (DMMs). A typical DMM can perform various measurements, including AC and DC voltage, AC and DC current, resistance, and continuity. For this experiment, the focus will be on measuring DC voltage and current.

4.3.5 Resistor Color Code

There are a variety of resistor in the course kit and while it's possible to measure all the values individually, learning how to read the color codes will make identifying individual resistors much easier. See the look up table below.



COLOR	1 ST BAND	2 ND BAND	3 RD BAND	MULTIPLIER	TOLERANCE
Black	0	0	0	1Ω	
Brown	1	1	1	10Ω	± 1% (F)
Red	2	2	2	100Ω	± 2% (G)
Orange	3	3	3	1KΩ	
Yellow	4	4	4	10KΩ	
Green	5	5	5	100KΩ	± 0.5% (D)
Blue	6	6	6	1MΩ	± 0.25% (C)
Violet	7	7	7	10MΩ	± 0.10% (B)
Grey	8	8	8		± 0.05%
White	9	9	9		
Gold				0.1Ω	± 5% (J)
Silver				0.01Ω	± 10% (K)

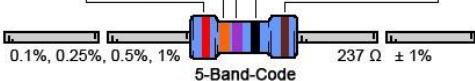


Figure 4.3: Resistor color code look up table.

4.4 Big Picture

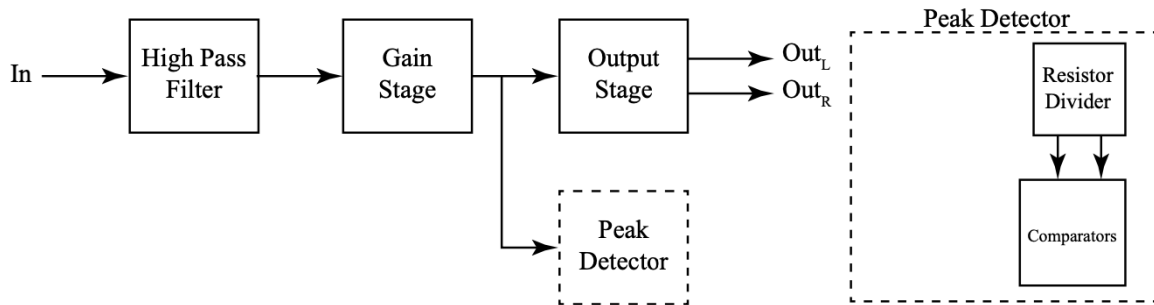


Figure 4.4: Big picture with emphasis on the peak detector.

This lab focuses on basic resistor circuits which will help to form the resistor divider used to set the threshold voltages in the peak detector for the final project.

4.5 Pre-lab Requirements

Complete the following before coming to lab.

4.5.1 Resistor Ladders

1. Using the circuit in Figure 4.5, calculate the ideal voltages for nodes A, B, C, and D. Table the values for future comparison.
2. Construct the circuit in Figure 4.5 and measure the actual voltages using the oscilloscope. Table the values.

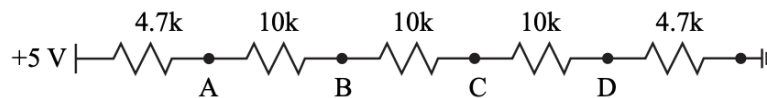


Figure 4.5: Resistor ladder example

4.5.2 Two-bit R-2R Ladder

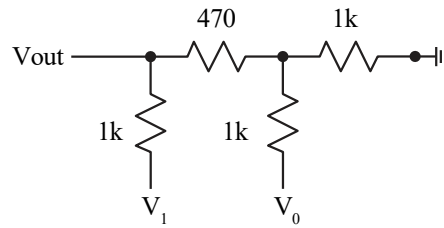


Figure 4.6: Two-bit R-2R ladder.

- Using the circuit in Figure 4.6, calculate the ideal output voltages for the following combination of voltages and table the values:
 - $V_0 = 0\text{ V}$ and $V_1 = 0\text{ V}$
 - $V_0 = 5\text{ V}$ and $V_1 = 0\text{ V}$
 - $V_0 = 0\text{ V}$ and $V_1 = 5\text{ V}$
 - $V_0 = 5\text{ V}$ and $V_1 = 5\text{ V}$
- Construct the circuit in Figure 4.6 and measure the output voltage using the oscilloscope and table the values. To generate the changes in voltages, set:
 - Wavegen 1 (V_0) to a 1 Hz square wave with a 2.5 V amplitude, a 2.5 V offset, and a 180 phase offset.
 - Wavegen 2 (V_1) to a 500 mHz square wave with a 2.5 V amplitude, a 2.5 V offset, and a 180 phase offset.
 - Make sure the Wavegen signals are set to synchronized.
- Save the circuit and Wavegen settings to demo to your lab instructor.
 - The output should look like the Scope image in Figure 4.7. In this circuit, V_0 and V_1 serve as digital inputs, switching between on (5 V) and off (0 V). As you “count” from 0 to 3 in binary, the resistor network creates proportional voltage steps at the output. This means that the output voltage represents the binary input as an analog voltage level between ground and 5 V, with each step increasing based on the weighted contributions of V_0 and V_1 .

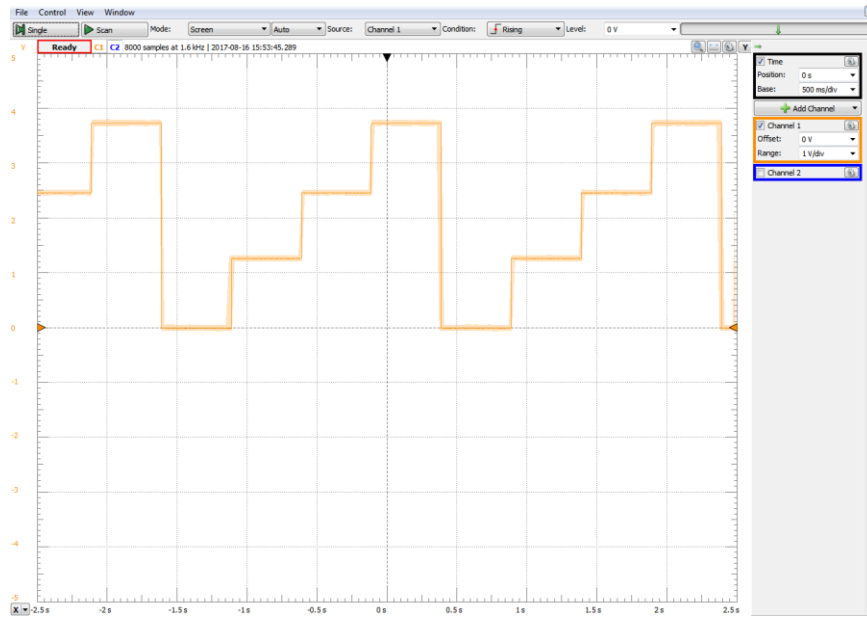


Figure 4.7: The desired output for the R-2R ladder circuit with the appropriate inputs.

4.6 In-Lab Requirements

The following items must be ready at the start of lab.

- Submit the table of voltages for the resistor ladder and the R-2R ladder to Canvas in a document.
- Demo the output of the R-2R ladder as V_0 and V_1 change.

4.6.1 Resistor Tolerances

There is no such thing as a perfect resistor. All resistors have tolerance about their specified value. For example, a $10\text{k}\Omega$ resistor with a $\pm 5\%$ tolerance will have a value between $10\text{k}\Omega - 5\% \leq R \leq 10\text{k}\Omega + 5\%$.

- Using the multimeter, measure the resistance of the resistors in the ladder from Subsection 4.5.1. Remember to remove the resistors from the breadboard so that they can be measured independently. Table these values.
- Re-calculate the ideal voltages using the measured resistances from the previous step and table the values.
- Using the multimeter, measure the resistance of the resistors used in the R-2R ladder from Subsection 4.5.2. Remember to remove the resistors from the breadboard so that they can be measured independently.

- Re-calculate the ideal voltages using the measured resistances from the previous step and table the values.

4.6.2 Three-bit R-2R ladder

Find someone in the lab to form a team of two. If there is an odd number of students, allow for a team of three where necessary. Complete the following:

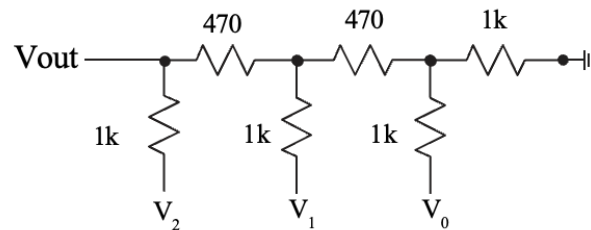


Figure 4.8: Three-bit R-2R ladder.

1. Construct the circuit in Figure 4.8 and measure the output voltage using the oscilloscope and table the values. To generate the changes in voltages, set:
 - (a) Wavegen 1 (V_0) to a 1 Hz square wave with a 2.5 V amplitude, a 2.5 V offset, and a 180 phase offset.
 - (b) Wavegen 2 (V_1) to a 500 mHz square wave with a 2.5 V amplitude, a 2.5 V offset, and a 180 phase offset.
 - (c) Make sure the Wavegen signals are set to synchronized.
 - (d) Since there is no third Wavegen wire, you will use the DAD's V+ wire (red) and ground wire (black) for V_2 .
2. Take a screenshot of the waveform of V_{out} , measure the values of each step, and record them in a table for the following cases:
 - (a) $V_2 = 0$ V.
 - (b) $V_2 = 5$ V.
3. Answer the questions listed in the Write up Section. Include the name of your teammate(s).

4.7 Write Up

Detail the effects of the resistance tolerances on the two different resistor ladders. Include the tabled values for both circuits and calculate the percent error using the experimentally measured values from the prelab and the values calculated from the measured resistors. You should have two tables: one for voltage values and their percent errors, and another for the resistor values and their percent errors.

Discuss the results of the teamwork section and answer the following:

- What did you learn from the teamwork section in terms of communication and collaboration?
- How would you rate your team's performance using a scale of 1 to 5, with 5 being the best? Discuss any errors your team might have had and how you corrected them.
- How is a three-bit ladder different than the two-bit ladder constructed in the prelab? What are some possible applications for the three-bit ladder?

For all lab write-up submissions and reports, the backgrounds in LTspice and Waveforms should be changed from dark to light to ensure readability for grading.

If the prelab simulation is incorrect, do not compare the in-lab results to wrong values. Correct numbers or circuit schematics should be used in the write-up report.